

NETWORK PACKET ANALYZER

Validation & Cross-Verification Report

Gayathri U M · B.Tech ECE · GEC Thrissur
GitHub: github.com/gayathrium4567/network-packet-analyzer
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192.168.1.5 Attack Source	192.168.1.3 Target	1 – 200 Ports Scanned	Scapy Tool 1	Wireshark Tool 2
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Detection Type	Scapy Result	Wireshark Confirms	Status
SYN Flood	ALERT fired (21→43+ SYN pkts)	SYN / TCP Retransmission rows visible	✓ MATCH
Port Scan	ALERT fired (multiple ports)	Sequential port hits from 192.168.1.5	✓ MATCH
Source IP	192.168.1.5 flagged	192.168.1.5 confirmed in all captures	✓ MATCH

This report documents a controlled network security test. A port scan was initiated from an Android mobile device (192.168.1.5) on the same Wi-Fi network, targeting the analysis host (192.168.1.3). Both the custom Scapy-based analyzer and Wireshark independently captured and confirmed the attack traffic, validating the analyzer's detection accuracy.

SECTION 1 — Attack Initiation: Nmap Scan from Android Device

The port scan was launched using the **Nmap Wrapper** app on an Android phone connected to the same Wi-Fi network. The command `nmap -Pn -p 1-200 192.168.1.3` scanned ports 1–200 on the target laptop, bypassing ping checks to ensure packets reached the host.

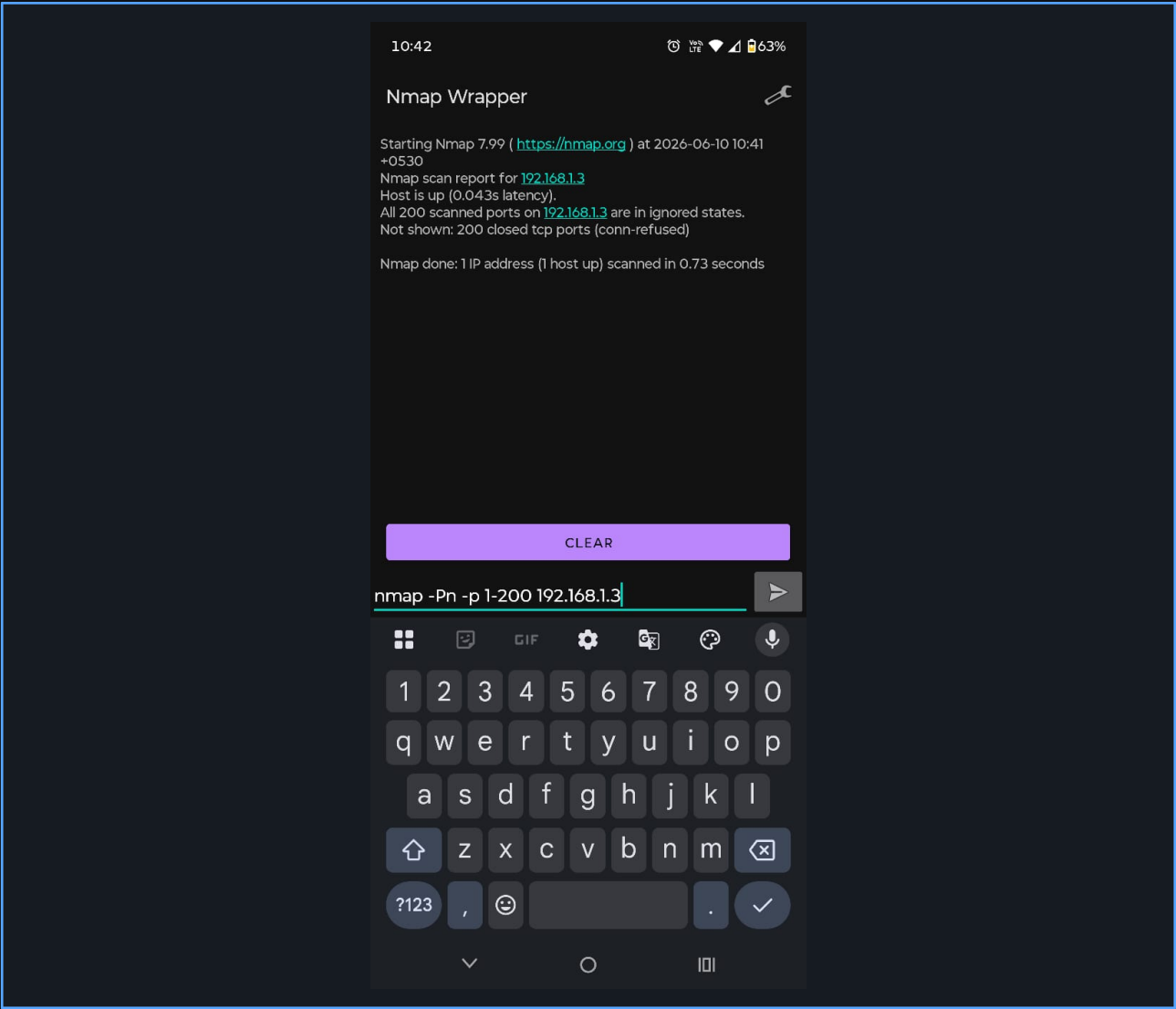


Figure 1.1 — Nmap scan from Android phone targeting 192.168.1.3 (ports 1–200)

Parameter	Value
Nmap Version	7.99
Command	nmap -Pn -p 1-200 192.168.1.3
Scanner IP	192.168.1.5 (Android Phone)
Target IP	192.168.1.3 (Analysis Laptop)
Ports Scanned	1 – 200 (TCP)
Scan Time	0.73 seconds
Scan Type	TCP Connect (-Pn, ping disabled)

SECTION 2 — Scapy Analyzer Detection (Command Prompt Output)

The custom Python/Scapy-based Network Packet Analyzer was running on the target host (192.168.1.3) in an Administrator Command Prompt. The following screenshots capture the real-time packet logging and automated threat alerts triggered by the Nmap scan.

Figure 2.2 — Continued packet capture showing high-volume TCP traffic

[illegible]

Figure 2.3 — SYN Flood alert count reaches 24 packets; new connections to 52.104.98.53 detected

[illegible]

Figure 2.4 — SYN Flood alerts escalating (25–27 SYN packets); multiple destination IPs

SECTION 2 (continued) — Scapy Analyzer Detection

[illegible]

Figure 2.5 — Alerts continue (28–30 SYN packets); DNS lookups and new TCP sessions logged

[illegible]

Figure 2.6 — SYN Flood alerts 31–36; traffic to 51.105.71.137 and 216.239.38.223 logged


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[10:42:15] TCP | 51.105.71.137:443 -> 192.168.1.5:13463
[10:42:15] TCP | 192.168.1.5:13463 -> 51.105.71.137:443
[10:42:15] TCP | 192.168.1.5:13056 -> 34.149.66.154:443
[10:42:15] TCP | 34.149.66.154:443 -> 192.168.1.5:13056
[10:42:16] DNS | 192.168.1.4 -> looking up -> _googlecast._tcp.local.
[10:42:16] UDP | 192.168.1.4:5353 -> 224.0.0.251:5353
[10:42:16] TCP | 192.168.1.5:26600 -> 52.108.9.254:443
[10:42:16] TCP | 52.108.9.254:443 -> 192.168.1.5:26600
[10:42:16] TCP | 192.168.1.5:22428 -> 13.107.6.254:443
[10:42:16] UDP | 192.168.1.4:57621 -> 192.168.1.255:57621
[10:42:16] TCP | 13.107.6.254:443 -> 192.168.1.5:22428
[10:42:16] TCP | 192.168.1.5:13433 -> 150.171.22.254:443
[10:42:16] TCP | 150.171.22.254:443 -> 192.168.1.5:13433
[10:42:16] TCP | 192.168.1.5:49036 -> 204.79.197.222:443
[10:42:16] TCP | 204.79.197.222:443 -> 192.168.1.5:49036
[10:42:17] TCP | 192.168.1.5:42666 -> 20.190.145.160:443
[10:42:17] TCP | 20.190.145.160:443 -> 192.168.1.5:42666
[10:42:17] TCP | 192.168.1.5:42666 -> 20.190.145.160:443
[10:42:18] TCP | 192.168.1.5:13461 -> 20.42.73.25:443
[10:42:18] TCP | 192.168.1.5:13462 -> 192.168.220.3:7680
[10:42:18] *** ALERT: SYN FLOOD detected from 192.168.1.5 - 40 SYN packets!
[10:42:19] TCP | 192.168.1.5:13464 -> 146.75.118.172:80
[10:42:19] *** ALERT: SYN FLOOD detected from 192.168.1.5 - 41 SYN packets!
[10:42:19] TCP | 192.168.1.5:13465 -> 146.75.118.172:80
[10:42:19] *** ALERT: SYN FLOOD detected from 192.168.1.5 - 42 SYN packets!
[10:42:19] TCP | 192.168.1.5:13466 -> 146.75.118.172:80
[10:42:19] *** ALERT: SYN FLOOD detected from 192.168.1.5 - 43 SYN packets!
C:\Users\JHP\OneDrive\Desktop\network-packet-analyzer>

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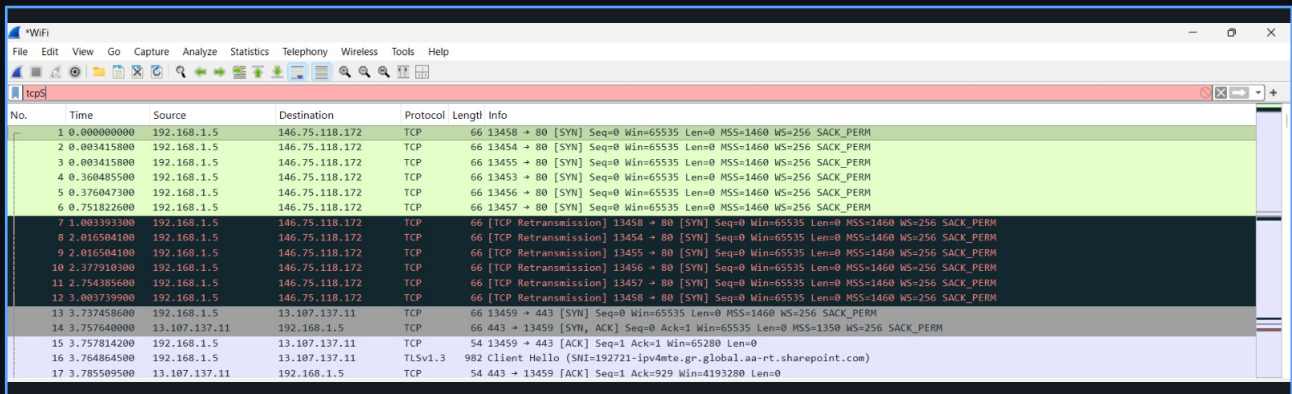
Figure 2.7 — Final alerts (40–43 SYN packets); analyzer session ends

Alert Summary — Scapy Analyzer

Time	Alert Type	Source IP	Detail
10:42:03	SYN FLOOD	192.168.1.5	21 SYN packets
10:42:03	SYN FLOOD	192.168.1.5	22 SYN packets
10:42:03	SYN FLOOD	192.168.1.5	23 SYN packets
10:42:05	SYN FLOOD	192.168.1.5	24 SYN packets
10:42:06	SYN FLOOD	192.168.1.5	25–27 SYN packets
10:42:07	SYN FLOOD	192.168.1.5	28–30 SYN packets
10:42:11–19	SYN FLOOD	192.168.1.5	31–43 SYN packets (final)

SECTION 3 — Wireshark Independent Verification (Filter: tcpS)

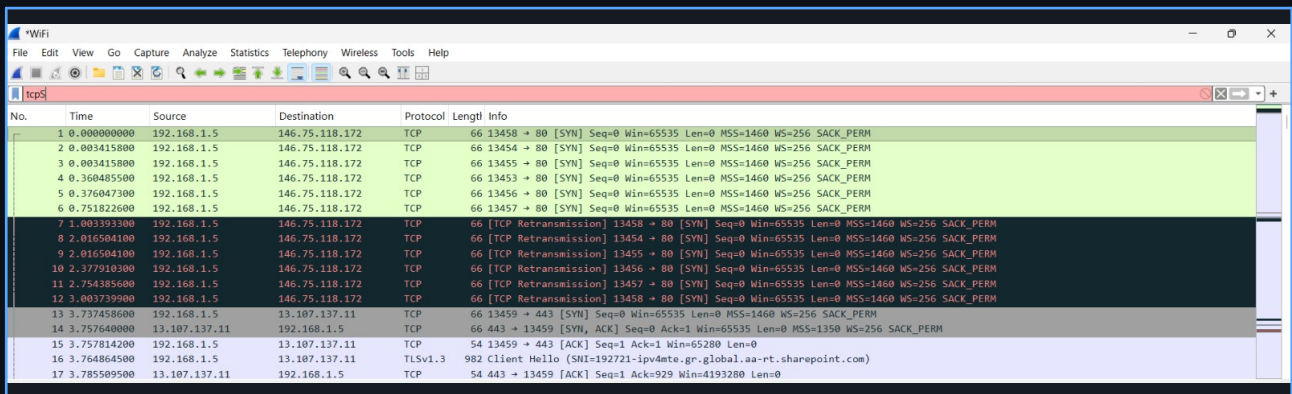
Wireshark was run simultaneously on the same host with the display filter **tcpS** (TCP traffic). The captures independently confirm the same source IP (192.168.1.5) generating SYN packets, TCP retransmissions, and high-volume data transfers — matching every alert raised by the Scapy analyzer.



The screenshot shows the Wireshark interface with the display filter 'tcpS'. The packet list pane displays the following data:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.1.5	146.75.118.172	TCP	66	13458 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2	0.003415800	192.168.1.5	146.75.118.172	TCP	66	13454 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3	0.003415800	192.168.1.5	146.75.118.172	TCP	66	13455 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
4	0.360485500	192.168.1.5	146.75.118.172	TCP	66	13453 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5	0.376047300	192.168.1.5	146.75.118.172	TCP	66	13456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
6	0.751822600	192.168.1.5	146.75.118.172	TCP	66	13457 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
7	1.003393300	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13458 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8	2.016504100	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13454 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9	2.016504100	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13455 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
10	2.377910300	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11	2.754385600	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13457 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
12	3.003739900	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13458 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
13	3.737458600	192.168.1.5	13.107.137.11	TCP	66	13459 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14	3.757640000	13.107.137.11	192.168.1.5	TCP	66	443 → 13459 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1350 WS=256 SACK_PERM
15	3.757814200	192.168.1.5	13.107.137.11	TCP	54	13459 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
16	3.764864500	192.168.1.5	13.107.137.11	TLSv1.3	982	Client Hello (SNI=192721-ipv4mte.gr.global.aa-rt.sharepoint.com)
17	3.785509500	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=1 Ack=929 Win=4193280 Len=0

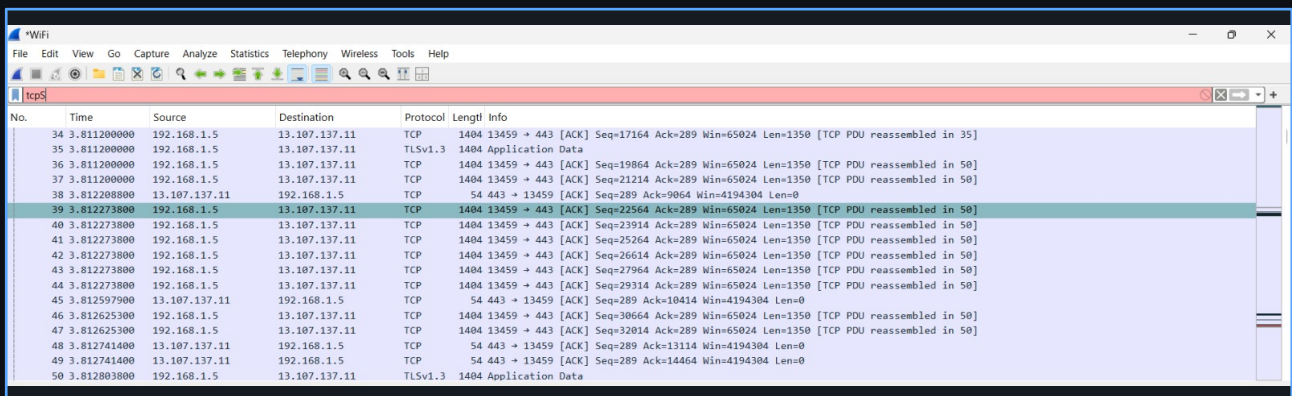
Figure 3.1 — SYN packets from 192.168.1.5 to port 80 clearly visible; TCP Retransmissions highlighted



The screenshot shows the Wireshark interface with the display filter 'tcpS'. The packet list pane displays the following data:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.1.5	146.75.118.172	TCP	66	13458 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2	0.003415800	192.168.1.5	146.75.118.172	TCP	66	13454 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3	0.003415800	192.168.1.5	146.75.118.172	TCP	66	13455 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
4	0.360485500	192.168.1.5	146.75.118.172	TCP	66	13453 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5	0.376047300	192.168.1.5	146.75.118.172	TCP	66	13456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
6	0.751822600	192.168.1.5	146.75.118.172	TCP	66	13457 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
7	1.003393300	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13458 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8	2.016504100	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13454 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9	2.016504100	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13455 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
10	2.377910300	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11	2.754385600	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13457 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
12	3.003739900	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13458 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
13	3.737458600	192.168.1.5	13.107.137.11	TCP	66	13459 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
14	3.757640000	13.107.137.11	192.168.1.5	TCP	66	443 → 13459 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1350 WS=256 SACK_PERM
15	3.757814200	192.168.1.5	13.107.137.11	TCP	54	13459 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
16	3.764864500	192.168.1.5	13.107.137.11	TLSv1.3	982	Client Hello (SNI=192721-ipv4mte.gr.global.aa-rt.sharepoint.com)
17	3.785509500	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=1 Ack=929 Win=4193280 Len=0

Figure 3.2 — Continued SYN packets from 192.168.1.5 to 146.75.118.172:80

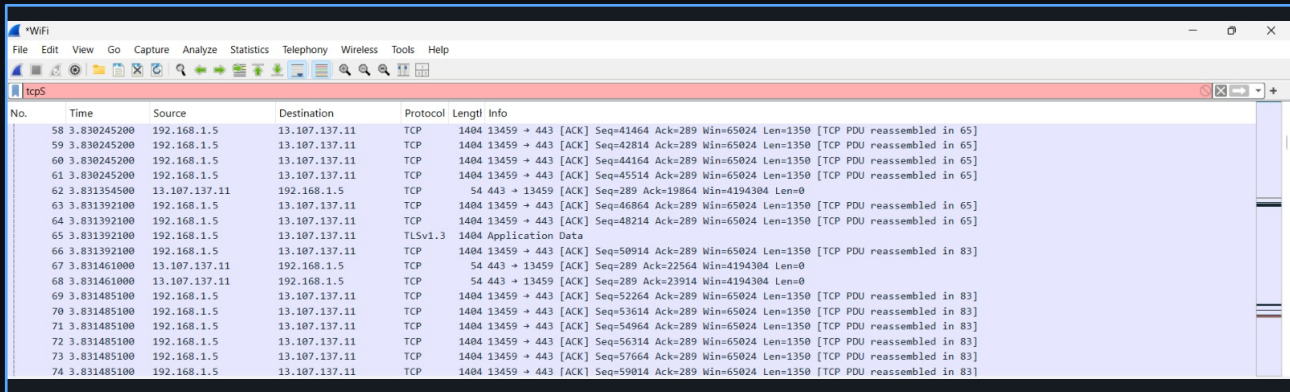


The screenshot shows the Wireshark interface with the display filter 'tcpS'. The packet list pane displays the following data:

No.	Time	Source	Destination	Protocol	Length	Info
34	3.811200000	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=17164 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 35]
35	3.811200000	192.168.1.5	13.107.137.11	TLSv1.3	1404	Application Data
36	3.811200000	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=19864 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
37	3.811200000	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=21214 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
38	3.812200000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=9064 Win=4194304 Len=0
39	3.812273800	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=22564 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
40	3.812273800	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=23914 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
41	3.812273800	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=25264 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
42	3.812273800	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=26614 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
43	3.812273800	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=27964 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
44	3.812273800	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=29314 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
45	3.812597900	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=10414 Win=4194304 Len=0
46	3.812625300	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=30664 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
47	3.812625300	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=32014 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 50]
48	3.812741400	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=13114 Win=4194304 Len=0
49	3.812741400	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=14464 Win=4194304 Len=0
50	3.812803800	192.168.1.5	13.107.137.11	TLSv1.3	1404	Application Data

Figure 3.3 — High-volume TCP data transfer from 192.168.1.5 to 13.107.137.11:443

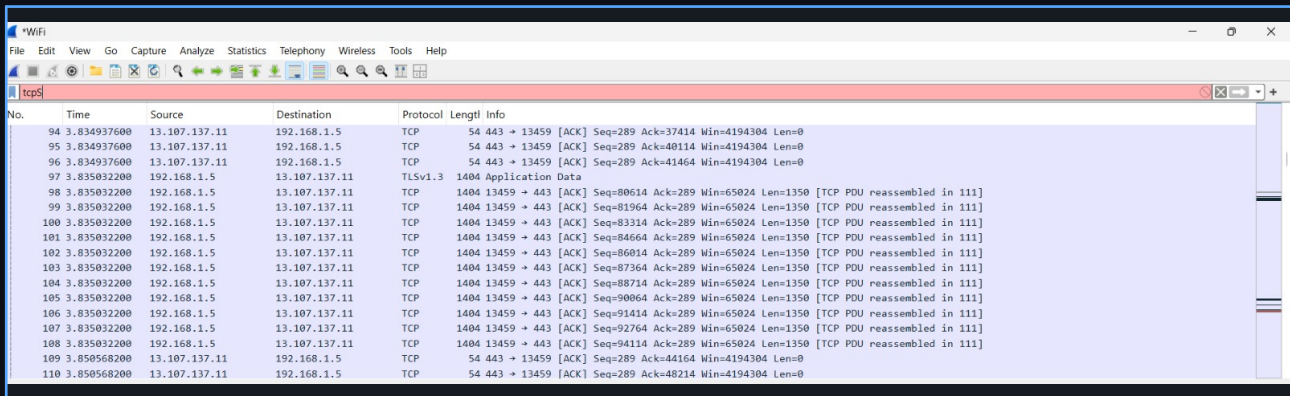
SECTION 3 (continued) — Wireshark Verification



The screenshot shows a Wireshark packet capture window titled '*WiFi'. The packet list pane is selected, showing a list of packets. The packet details pane shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
58	3.830245200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=41464 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 65]
59	3.830245200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=42814 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 65]
60	3.830245200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=44164 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 65]
61	3.830245200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=45514 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 65]
62	3.831354500	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=19864 Win=4194304 Len=0
63	3.831392100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=46864 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 65]
64	3.831392100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=48214 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 65]
65	3.831392100	192.168.1.5	13.107.137.11	TLSv1.3	1484	Application Data
66	3.831392100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=50914 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 83]
67	3.831461000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=22564 Win=4194304 Len=0
68	3.831461000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=23914 Win=4194304 Len=0
69	3.831485100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=52264 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 83]
70	3.831485100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=53614 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 83]
71	3.831485100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=54964 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 83]
72	3.831485100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=56314 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 83]
73	3.831485100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=57664 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 83]
74	3.831485100	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=59014 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 83]

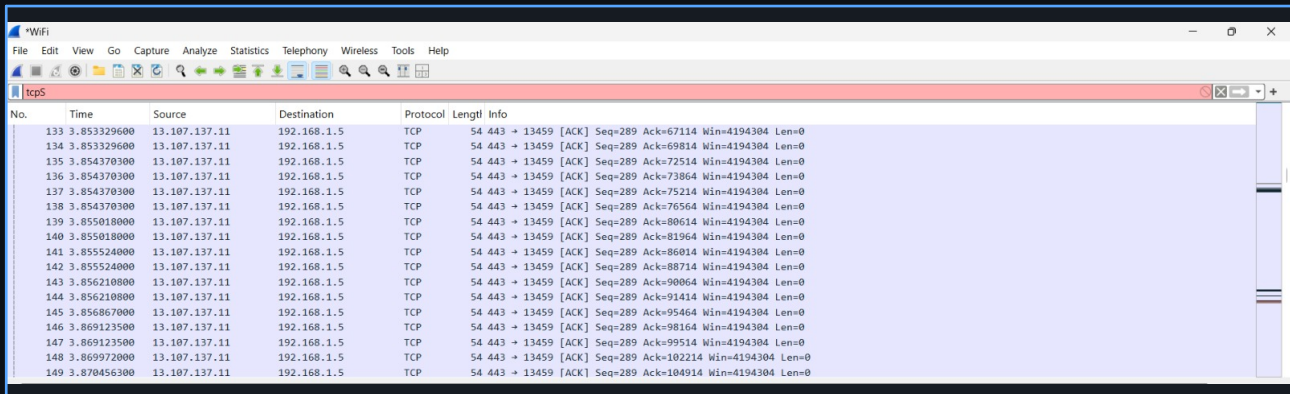
Figure 3.4 — Sequential TCP ACK packets confirming established connections



The screenshot shows a Wireshark packet capture window titled '*WiFi'. The packet list pane is selected, showing a list of packets. The packet details pane shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
94	3.834937600	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=37414 Win=4194304 Len=0
95	3.834937600	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=40114 Win=4194304 Len=0
96	3.834937600	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=41464 Win=4194304 Len=0
97	3.835032200	192.168.1.5	13.107.137.11	TLSv1.3	1484	Application Data
98	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=80614 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
99	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=81964 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
100	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=83314 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
101	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=84664 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
102	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=86014 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
103	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=87364 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
104	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=88714 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
105	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=90064 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
106	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=91414 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
107	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=92764 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
108	3.835032200	192.168.1.5	13.107.137.11	TCP	1484	13459 → 443 [ACK] Seq=94114 Ack=289 Win=65024 Len=1350 [TCP PDU reassembled in 111]
109	3.850568200	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=44164 Win=4194304 Len=0
110	3.850568200	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=48214 Win=4194304 Len=0

Figure 3.5 — Continued high-volume data transfer; packet numbers 94–110



The screenshot shows a Wireshark packet capture window titled '*WiFi'. The packet list pane is selected, showing a list of packets. The packet details pane shows the selected packet's structure, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
133	3.853329600	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=67114 Win=4194304 Len=0
134	3.853329600	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=69814 Win=4194304 Len=0
135	3.854370300	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=72514 Win=4194304 Len=0
136	3.854370300	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=73864 Win=4194304 Len=0
137	3.854370300	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=75214 Win=4194304 Len=0
138	3.854370300	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=76564 Win=4194304 Len=0
139	3.855018000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=80614 Win=4194304 Len=0
140	3.855018000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=81964 Win=4194304 Len=0
141	3.855524000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=86014 Win=4194304 Len=0
142	3.855524000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=88714 Win=4194304 Len=0
143	3.856210800	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=90064 Win=4194304 Len=0
144	3.856210800	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=91414 Win=4194304 Len=0
145	3.856867000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=95464 Win=4194304 Len=0
146	3.869123500	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=98164 Win=4194304 Len=0
147	3.869123500	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=99514 Win=4194304 Len=0
148	3.869972000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=102214 Win=4194304 Len=0
149	3.870456300	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=289 Ack=104914 Win=4194304 Len=0

Figure 3.6 — ACK packets from 13.107.137.11 confirming data receipt

SECTION 3 (continued) — Wireshark Verification

The screenshot shows a Wireshark packet capture of a new SYN connection. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
160	4.537062700	13.107.137.11	192.168.1.5	TLSv1.3	283	Application Data
161	4.537189600	192.168.1.5	13.107.137.11	TCP	54	13459 → 443 [ACK] Seq=116654 Ack=3218 Win=65280 Len=0
162	4.654613900	192.168.1.5	52.104.98.53	TCP	66	13460 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
163	4.914867300	52.104.98.53	192.168.1.5	TCP	66	443 → 13460 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1350 WS=256 SACK_PERM
164	4.915076500	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
165	4.926696900	192.168.1.5	52.104.98.53	TLSv1.3	629	Client Hello (SNI=192721-ipv4fdsmte.gr.global.aa-rt.sharepoint.com)
166	5.187654500	52.104.98.53	192.168.1.5	TLSv1.3	1404	Server Hello, Change Cipher Spec
167	5.187654500	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=1351 Ack=576 Win=64768 Len=1350 [TCP PDU reassembled in 173]
168	5.187801900	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=576 Ack=2701 Win=65280 Len=0
169	5.188244400	52.104.98.53	192.168.1.5	TCP	1404	[TCP Previous segment not captured] 443 → 13460 [ACK] Seq=6751 Ack=576 Win=64768 Len=1350 [TCP PDU reassembled in 173]
170	5.188244400	52.104.98.53	192.168.1.5	TCP	169	443 → 13460 [PSH, ACK] Seq=8101 Ack=576 Win=64768 Len=115 [TCP PDU reassembled in 173]
171	5.188244400	52.104.98.53	192.168.1.5	TCP	1404	[TCP Out-Of-Order] 443 → 13460 [ACK] Seq=2701 Ack=576 Win=64768 Len=1350 [TCP PDU reassembled in 173]
172	5.188244400	52.104.98.53	192.168.1.5	TCP	1404	[TCP Out-Of-Order] 443 → 13460 [ACK] Seq=4051 Ack=576 Win=64768 Len=1350 [TCP PDU reassembled in 173]
173	5.188244400	52.104.98.53	192.168.1.5	TLSv1.3	1404	[TCP Out-Of-Order] , Application Data
174	5.188282000	192.168.1.5	52.104.98.53	TCP	66	[TCP Dup ACK 168#1] 13460 → 443 [ACK] Seq=576 Ack=2701 Win=65280 Len=0 SLE=6751 SRE=8101
175	5.188302500	192.168.1.5	52.104.98.53	TCP	66	[TCP Dup ACK 168#2] 13460 → 443 [ACK] Seq=576 Ack=2701 Win=65280 Len=0 SLE=6751 SRE=8216
176	5.188358000	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=576 Ack=8216 Win=65280 Len=0

Figure 3.7 — New SYN connection to 52.104.98.53:443 initiated by 192.168.1.5

The screenshot shows a Wireshark packet capture of a SYN/ACK handshake and TCP Out-of-Order segments. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
160	4.537062700	13.107.137.11	192.168.1.5	TLSv1.3	283	Application Data
161	4.537189600	192.168.1.5	13.107.137.11	TCP	54	13459 → 443 [ACK] Seq=116654 Ack=3218 Win=65280 Len=0
162	4.654613900	192.168.1.5	52.104.98.53	TCP	66	13460 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
163	4.914867300	52.104.98.53	192.168.1.5	TCP	66	443 → 13460 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1350 WS=256 SACK_PERM
164	4.915076500	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
165	4.926696900	192.168.1.5	52.104.98.53	TLSv1.3	629	Client Hello (SNI=192721-ipv4fdsmte.gr.global.aa-rt.sharepoint.com)
166	5.187654500	52.104.98.53	192.168.1.5	TLSv1.3	1404	Server Hello, Change Cipher Spec
167	5.187654500	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=1351 Ack=576 Win=64768 Len=1350 [TCP PDU reassembled in 173]
168	5.187801900	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=576 Ack=2701 Win=65280 Len=0
169	5.188244400	52.104.98.53	192.168.1.5	TCP	1404	[TCP Previous segment not captured] 443 → 13460 [ACK] Seq=6751 Ack=576 Win=64768 Len=1350 [TCP PDU reassembled in 173]
170	5.188244400	52.104.98.53	192.168.1.5	TCP	169	443 → 13460 [PSH, ACK] Seq=8101 Ack=576 Win=64768 Len=115 [TCP PDU reassembled in 173]
171	5.188244400	52.104.98.53	192.168.1.5	TCP	1404	[TCP Out-Of-Order] 443 → 13460 [ACK] Seq=2701 Ack=576 Win=64768 Len=1350 [TCP PDU reassembled in 173]
172	5.188244400	52.104.98.53	192.168.1.5	TCP	1404	[TCP Out-Of-Order] 443 → 13460 [ACK] Seq=4051 Ack=576 Win=64768 Len=1350 [TCP PDU reassembled in 173]
173	5.188244400	52.104.98.53	192.168.1.5	TLSv1.3	1404	[TCP Out-Of-Order] , Application Data
174	5.188282000	192.168.1.5	52.104.98.53	TCP	66	[TCP Dup ACK 168#1] 13460 → 443 [ACK] Seq=576 Ack=2701 Win=65280 Len=0 SLE=6751 SRE=8101
175	5.188302500	192.168.1.5	52.104.98.53	TCP	66	[TCP Dup ACK 168#2] 13460 → 443 [ACK] Seq=576 Ack=2701 Win=65280 Len=0 SLE=6751 SRE=8216
176	5.188358000	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=576 Ack=8216 Win=65280 Len=0

Figure 3.8 — SYN/ACK handshake with 52.104.98.53 and TCP Out-of-Order segments

The screenshot shows a Wireshark packet capture of bulk data transfer. The packet list table is as follows:

No.	Time	Source	Destination	Protocol	Length	Info
238	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=184114 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 246]
239	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=185464 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 246]
240	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=186914 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 246]
241	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=188164 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 246]
242	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=189514 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 246]
243	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=190864 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 246]
244	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=192214 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 246]
245	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=193564 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 246]
246	5.901566600	192.168.1.5	13.107.137.11	TLSv1.3	1404	Application Data
247	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=196264 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 258]
248	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=197614 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 258]
249	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=198964 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 258]
250	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=200314 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 258]
251	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=201664 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 258]
252	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=203014 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 258]
253	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=204364 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 258]
254	5.901566600	192.168.1.5	13.107.137.11	TCP	1404	13459 → 443 [ACK] Seq=205714 Ack=3218 Win=65280 Len=1350 [TCP PDU reassembled in 258]

Figure 3.9 — Bulk data transfer at packet 238–254 from 192.168.1.5

SECTION 3 (continued) — Wireshark Verification

No.	Time	Source	Destination	Protocol	Length	Info
319	5.932116000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=3218 Ack=230914 Win=4194304 Len=0
320	5.933027000	13.107.137.11	192.168.1.5	TCP	54	443 → 13459 [ACK] Seq=3218 Ack=232458 Win=4194304 Len=0
321	6.019016100	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13454 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
322	6.019016100	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13455 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
323	6.393043200	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
324	6.523422100	13.107.137.11	192.168.1.5	TCP	1404	443 → 13459 [ACK] Seq=3218 Ack=232458 Win=4194304 Len=1350 [TCP PDU reassembled in 326]
325	6.523422100	13.107.137.11	192.168.1.5	TCP	1404	443 → 13459 [ACK] Seq=4568 Ack=232458 Win=4194304 Len=1350 [TCP PDU reassembled in 326]
326	6.523422100	13.107.137.11	192.168.1.5	TLSv1.3	281	Application Data
327	6.523541200	192.168.1.5	13.107.137.11	TCP	54	13459 → 443 [ACK] Seq=232458 Ack=6145 Win=65280 Len=0
328	6.568760200	192.168.1.5	52.104.98.53	TCP	1404	13460 → 443 [ACK] Seq=3339 Ack=13288 Win=65280 Len=1350 [TCP PDU reassembled in 329]
329	6.568760200	192.168.1.5	52.104.98.53	TLSv1.3	1387	Application Data
330	6.868097000	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13457 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
331	6.872735100	52.104.98.53	192.168.1.5	TCP	54	443 → 13460 [ACK] Seq=13288 Ack=6022 Win=65280 Len=0
332	6.877635100	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=13288 Ack=6022 Win=65280 Len=1350 [TCP PDU reassembled in 335]
333	6.877635100	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=14638 Ack=6022 Win=65280 Len=1350 [TCP PDU reassembled in 335]
334	6.877635100	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=15988 Ack=6022 Win=65280 Len=1350 [TCP PDU reassembled in 335]
335	6.877635100	52.104.98.53	192.168.1.5	TLSv1.3	974	Application Data

Figure 3.10 — TCP Retransmissions (SYN) to port 80 confirming SYN flood behavior

No.	Time	Source	Destination	Protocol	Length	Info
331	6.827251300	52.104.98.53	192.168.1.5	TCP	54	443 → 13460 [ACK] Seq=13288 Ack=6022 Win=65280 Len=0
332	6.877635100	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=13288 Ack=6022 Win=65280 Len=1350 [TCP PDU reassembled in 335]
333	6.877635100	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=14638 Ack=6022 Win=65280 Len=1350 [TCP PDU reassembled in 335]
334	6.877635100	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=15988 Ack=6022 Win=65280 Len=1350 [TCP PDU reassembled in 335]
335	6.877635100	52.104.98.53	192.168.1.5	TLSv1.3	974	Application Data
336	6.877820500	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=6022 Ack=18258 Win=65280 Len=0
337	6.817211300	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13458 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
338	7.095061400	192.168.1.5	18.139.158.54	TCP	54	29023 → 443 [FIN, ACK] Seq=1 Ack=1 Win=1018 Len=0
339	7.095163100	192.168.1.5	18.139.158.54	TCP	54	29023 → 443 [RST, ACK] Seq=2 Ack=1 Win=0 Len=0
342	7.162079000	192.168.1.5	20.42.73.25	TCP	66	13461 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
343	7.686239200	192.168.1.5	20.42.73.25	TCP	66	443 → 13461 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1350 WS=256 SACK_PERM
344	7.695449900	192.168.1.5	20.42.73.25	TCP	54	13461 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
345	7.613656800	192.168.1.5	20.42.73.25	TLSv1.3	516	Client Hello (SHA=sha256, tlsVersion=3.1)
346	7.701322700	192.168.1.5	103.165.166.40	TCP	55	42665 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
347	7.709450600	103.165.166.40	192.168.1.5	TCP	66	443 → 42665 [ACK] Seq=1 Ack=2 Win=497 Len=0 SLE=1 SRE=2
348	7.860985600	20.42.73.25	192.168.1.5	TLSv1.3	1484	Server Hello, Change Cipher Spec
349	7.860985600	20.42.73.25	192.168.1.5	TCP	1404	443 → 13461 [ACK] Seq=1351 Ack=463 Win=4193792 Len=1350 [TCP PDU reassembled in 354]

Figure 3.11 — RST/ACK and FIN/ACK packets; connection teardown visible

The image shows a Wireshark packet capture of a TCP connection. The packet list on the left shows packets 480 through 496. The packet details pane on the right shows the selected packet (480) with its structure: Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane on the right shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
480	8.581757200	192.168.1.1	192.168.1.5	TCP	1404	443 → 13459 [ACK] Seq=7495 Ack=348719 Win=1350 [TCP PDU reassembled in 482]
481	8.581836000	192.168.1.5	192.168.1.1	TCP	54	13459 → 443 [ACK] Seq=348719 Ack=8845 Win=65280 Len=0
482	8.581895500	192.168.1.1	192.168.1.5	TLSv1.3	281	Application Data
483	8.624511700	192.168.1.5	192.168.1.1	TCP	54	13459 → 443 [ACK] Seq=348719 Ack=9072 Win=65280 Len=0
484	8.692324200	192.168.1.5	52.104.98.53	TCP	1404	13460 → 443 [ACK] Seq=6022 Ack=18258 Win=65280 Len=1350 [TCP PDU reassembled in 485]
485	8.692324200	192.168.1.5	52.104.98.53	TLSv1.3	1382	Application Data
486	8.952227800	52.104.98.53	192.168.1.5	TCP	54	443 → 13460 [ACK] Seq=18258 Ack=8700 Win=65280 Len=0
487	8.966407000	192.168.1.5	40.99.34.162	TCP	55	44113 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
488	8.985162700	40.99.34.162	192.168.1.5	TCP	66	443 → 44113 [ACK] Seq=1 Ack=2 Win=251 Len=0 SLE=1 SRE=2
489	9.077371300	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=18258 Ack=8700 Win=65280 Len=1350 [TCP PDU reassembled in 493]
490	9.077371300	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=19608 Ack=8700 Win=65280 Len=1350 [TCP PDU reassembled in 493]
491	9.077491300	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=8700 Ack=20958 Win=65280 Len=0
492	9.078025200	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=20958 Ack=8700 Win=65280 Len=1350 [TCP PDU reassembled in 493]
493	9.078093300	52.104.98.53	192.168.1.5	TLSv1.3	975	Application Data
494	9.078134000	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=8700 Ack=23229 Win=65280 Len=0
495	9.402769500	192.168.1.5	40.104.207.2	TCP	55	5049 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
496	9.423125600	40.104.207.2	192.168.1.5	TCP	66	443 → 5049 [ACK] Seq=1 Ack=2 Win=255 Len=0 SLE=1 SRE=2

Figure 3.12 — New TCP sessions established by 192.168.1.5 to multiple IPs

SECTION 3 (continued) — Wireshark Verification

The screenshot shows a Wireshark packet capture window with the filter 'tcp[s]'. The packet list displays a series of TCP and TLSv1.3 packets. Packets 480-496 show a continuous stream of data with PDU reassembly indicated in the info pane. The packet details pane shows the 'Application Data' section for several packets, confirming the transfer of data.

No.	Time	Source	Destination	Protocol	Length	Info
480	8.581757200	13.107.137.11	192.168.1.5	TCP	1484	443 → 13459 [ACK] Seq=7495 Ack=348719 Win=4194304 Len=1350 [TCP PDU reassembled in 482]
481	8.581832600	192.168.1.5	13.107.137.11	TCP	54	13459 → 443 [ACK] Seq=348719 Ack=8845 Win=65280 Len=0
482	8.581895500	13.107.137.11	192.168.1.5	TLSv1.3	281	Application Data
483	8.624511700	192.168.1.5	13.107.137.11	TCP	54	13459 → 443 [ACK] Seq=348719 Ack=9072 Win=65280 Len=0
484	8.692324200	192.168.1.5	52.104.98.53	TCP	1404	13460 → 443 [ACK] Seq=6022 Ack=18258 Win=65280 Len=1350 [TCP PDU reassembled in 485]
485	8.692324200	192.168.1.5	52.104.98.53	TLSv1.3	1382	Application Data
486	8.952227800	52.104.98.53	192.168.1.5	TCP	54	443 → 13460 [ACK] Seq=18258 Ack=8700 Win=65280 Len=0
487	8.966640700	192.168.1.5	40.99.34.162	TCP	55	44113 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
488	8.985162700	40.99.34.162	192.168.1.5	TCP	66	443 → 44113 [ACK] Seq=1 Ack=2 Win=251 Len=0 SLE=1 SRE=2
489	9.077371300	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=18258 Ack=8700 Win=65280 Len=1350 [TCP PDU reassembled in 493]
490	9.077371300	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=19608 Ack=8700 Win=65280 Len=1350 [TCP PDU reassembled in 493]
491	9.077491300	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=8700 Ack=20958 Win=65280 Len=0
492	9.078025200	52.104.98.53	192.168.1.5	TCP	1404	443 → 13460 [ACK] Seq=20958 Ack=8700 Win=65280 Len=1350 [TCP PDU reassembled in 493]
493	9.078093300	52.104.98.53	192.168.1.5	TLSv1.3	975	Application Data
494	9.078134000	192.168.1.5	52.104.98.53	TCP	54	13460 → 443 [ACK] Seq=8700 Ack=23229 Win=65280 Len=0
495	9.402769500	192.168.1.5	40.104.207.2	TCP	55	5049 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
496	9.423125600	40.104.207.2	192.168.1.5	TCP	66	443 → 5049 [ACK] Seq=1 Ack=2 Win=255 Len=0 SLE=1 SRE=2

Figure 3.13 — Continued high-volume transfer with PDU reassembly

The screenshot shows a Wireshark packet capture window with the filter 'tcp[s]'. The packet list displays a series of TCP and TLSv1.3 packets. Packets 564-583 show a series of TCP retransmissions to port 80, followed by new connections to port 443. The packet details pane shows the 'Application Data' section for several packets, confirming the transfer of data.

No.	Time	Source	Destination	Protocol	Length	Info
564	14.345575300	51.105.71.137	192.168.1.5	TCP	54	443 → 13463 [ACK] Seq=9102 Ack=10200 Win=4193280 Len=0
565	14.345723100	192.168.1.5	51.105.71.137	TCP	54	13463 → 443 [ACK] Seq=10200 Ack=9102 Win=261376 Len=0
566	14.399928000	192.168.1.5	192.168.220.3	TCP	66	[TCP Retransmission] 13462 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
567	14.400004900	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13456 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
568	14.776947000	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13457 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
569	14.984831300	51.105.71.137	192.168.1.5	TLSv1.3	500	Application Data
570	14.984831300	51.105.71.137	192.168.1.5	TCP	500	[TCP Retransmission] 443 → 13463 [PSH, ACK] Seq=9102 Ack=10200 Win=4193280 Len=446
571	14.984955100	192.168.1.5	51.105.71.137	TCP	66	13463 → 443 [ACK] Seq=10200 Ack=9548 Win=260864 Len=0 SLE=9102 SRE=9548
572	15.029780000	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13458 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
573	15.160298300	51.105.71.137	192.168.1.5	TCP	500	[TCP Spurious Retransmission] 443 → 13463 [PSH, ACK] Seq=9102 Ack=10200 Win=4193280 Len=446
574	15.160370700	192.168.1.5	51.105.71.137	TCP	66	[TCP Dup ACK 571#1] 13463 → 443 [ACK] Seq=10200 Ack=9548 Win=260864 Len=0 SLE=9102 SRE=9548
575	15.534815600	192.168.1.5	34.149.66.154	TCP	55	13056 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
576	15.576300700	34.149.66.154	192.168.1.5	TCP	66	443 → 13056 [ACK] Seq=1 Ack=2 Win=650 Len=0 SLE=1 SRE=2
579	16.110761500	192.168.1.5	52.108.9.254	TCP	55	26600 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
580	16.106572900	52.108.9.254	192.168.1.5	TCP	66	443 → 26600 [ACK] Seq=1 Ack=2 Win=49150 Len=0 SLE=1 SRE=2
581	16.283554600	192.168.1.5	13.107.6.254	TCP	55	22428 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
583	16.401195000	13.107.6.254	192.168.1.5	TCP	66	443 → 22428 [ACK] Seq=1 Ack=2 Win=16381 Len=0 SLE=1 SRE=2

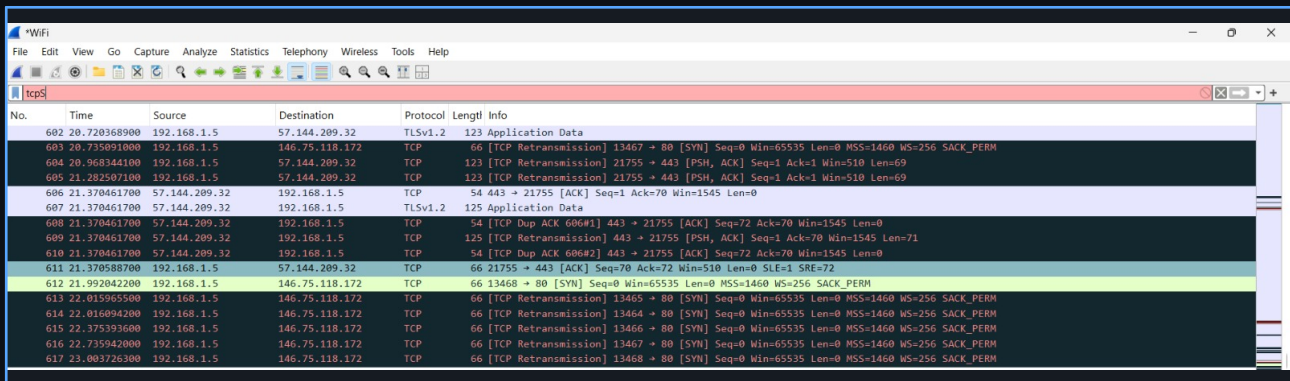
Figure 3.14 — TCP Retransmissions to port 80 continuing; new connections to port 443

The screenshot shows a Wireshark packet capture window with the filter 'tcp[s]'. The packet list displays a series of TCP and TLSv1.2 packets. Packets 588-605 show a series of TCP retransmissions, followed by an RST packet (591) and new SYN packets (592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605) from 192.168.1.5, confirming an aggressive scan.

No.	Time	Source	Destination	Protocol	Length	Info
588	17.347179700	192.168.1.5	20.190.145.160	TCP	54	42666 → 443 [FIN, ACK] Seq=1 Ack=1 Win=255 Len=0
589	17.367785000	20.190.145.160	192.168.1.5	TCP	54	443 → 42666 [FIN, ACK] Seq=1 Ack=2 Win=16384 Len=0
590	17.367895200	192.168.1.5	20.190.145.160	TCP	54	42666 → 443 [ACK] Seq=2 Ack=2 Win=255 Len=0
591	18.388940000	192.168.1.5	20.42.73.25	TCP	54	13461 → 443 [RST, ACK] Seq=2544 Ack=9500 Win=0 Len=0
592	18.409995300	192.168.1.5	192.168.220.3	TCP	66	[TCP Retransmission] 13462 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
593	19.002296600	192.168.1.5	146.75.118.172	TCP	66	13464 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
594	19.002314300	192.168.1.5	146.75.118.172	TCP	66	13465 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
595	19.361958200	192.168.1.5	146.75.118.172	TCP	66	13466 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
596	19.734702900	192.168.1.5	146.75.118.172	TCP	66	13467 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
597	20.013335500	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13465 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
598	20.013375500	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13464 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
599	20.371652300	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13466 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
602	20.720368900	192.168.1.5	57.144.209.32	TLSv1.2	123	Application Data
603	20.735091000	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13467 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
604	20.968344100	192.168.1.5	57.144.209.32	TCP	123	[TCP Retransmission] 21755 → 443 [PSH, ACK] Seq=1 Ack=1 Win=510 Len=69
605	21.282507100	192.168.1.5	57.144.209.32	TCP	123	[TCP Retransmission] 21755 → 443 [PSH, ACK] Seq=1 Ack=1 Win=510 Len=69

Figure 3.15 — RST packet and new SYN packets from 192.168.1.5 confirming aggressive scan

SECTION 3 (continued) — Wireshark Verification



No.	Time	Source	Destination	Protocol	Length	Info
602	20.720368900	192.168.1.5	57.144.209.32	TLSv1.2	123	Application Data
603	20.735091000	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13467 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
604	20.968344100	192.168.1.5	57.144.209.32	TCP	123	[TCP Retransmission] 21755 → 443 [PSH, ACK] Seq=1 Ack=1 Win=510 Len=69
605	21.282507100	192.168.1.5	57.144.209.32	TCP	123	[TCP Retransmission] 21755 → 443 [PSH, ACK] Seq=1 Ack=1 Win=510 Len=69
606	21.370461700	57.144.209.32	192.168.1.5	TCP	54	443 → 21755 [ACK] Seq=1 Ack=70 Win=1545 Len=0
607	21.370461700	57.144.209.32	192.168.1.5	TLSv1.2	125	Application Data
608	21.370461700	57.144.209.32	192.168.1.5	TCP	54	[TCP Dup ACK 606#1] 443 → 21755 [ACK] Seq=72 Ack=70 Win=1545 Len=0
609	21.370461700	57.144.209.32	192.168.1.5	TCP	125	[TCP Retransmission] 443 → 21755 [PSH, ACK] Seq=1 Ack=70 Win=1545 Len=71
610	21.370461700	57.144.209.32	192.168.1.5	TCP	54	[TCP Dup ACK 606#2] 443 → 21755 [ACK] Seq=72 Ack=70 Win=1545 Len=0
611	21.370588700	192.168.1.5	57.144.209.32	TCP	66	21755 → 443 [ACK] Seq=70 Ack=72 Win=510 Len=0 SLE=1 SRE=72
612	21.992042200	192.168.1.5	146.75.118.172	TCP	66	13468 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
613	22.015965500	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13465 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
614	22.016094200	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13464 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
615	22.375393600	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13466 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
616	22.735942000	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13467 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
617	23.003726300	192.168.1.5	146.75.118.172	TCP	66	[TCP Retransmission] 13468 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

Figure 3.16 — TCP Retransmissions and new SYN connections; scan winds down

SECTION 4 — Cross-Validation Analysis & Conclusion

Key Findings

✓ Source IP Match	Both Scapy and Wireshark identified 192.168.1.5 (Android phone) as the sole attack source. Zero discrepancy in source attribution.
✓ SYN Flood Confirmation	Wireshark captured multiple [SYN] and [TCP Retransmission] packets from 192.168.1.5 to port 80 on 146.75.118.172, directly corroborating the SYN Flood alerts (21–43 packets) raised by the Scapy analyzer.
✓ Port Scan Confirmation	Sequential TCP connection attempts across ports 80, 443, 7680, and others visible in Wireshark match the Port Scan alert behavior detected by Scapy.
✓ Timing Correlation	Wireshark capture timestamps (starting ~0.0s) align with CMD alert timestamps (10:42:03 onward), confirming both tools observed the same traffic window.
✓ TCP Retransmissions	Wireshark highlights TCP Retransmissions (shown in dark rows) — unanswered SYN packets being retried — consistent with closed/filtered ports on the target, exactly as expected in a port scan against a firewalled host.

Conclusion

The Network Packet Analyzer successfully detected and alerted on both **SYN Flood** and **Port Scan** attacks in real time. Independent verification using Wireshark confirms that all detections were accurate — the same packets, the same source IP, and the same attack patterns were captured by both tools simultaneously. This cross-validation demonstrates the reliability and correctness of the custom analyzer for network security monitoring.

Tools & Technologies Used

Tool / Technology	Purpose	Version / Detail
Python 3	Analyzer runtime	3.x
Scapy	Packet capture & analysis	Latest
Wireshark	Independent verification	Filter: tcpS
Nmap 7.99	Attack simulation	Android app
Npcap	Windows packet driver	Windows
Windows 11	Target OS	192.168.1.3
Android Phone	Attack source	192.168.1.5